Giovanni Maya

(213) 479-3389 | giovannimhern@uchicago.edu | linkedin.com/in/giomhern | github.com/giomhern

Education

University of Chicago | Chicago, IL *September 2022 – March 2027*

Bachelor of Science in Computer Science, Bachelor of Arts in Data Science

- **Relevant Coursework:** Theory of Algorithms, Discrete Mathematics, Computer Architecture for Scientists, Systems Programming I & II, Functional Programming, Usable Security and Privacy

Experience

Shopify Bellevue, WA

Data Engineering Intern *September 2026 – Present*

- Designed and began implementing an extensible **data-quality system** for upstream schema, volume, and distribution anomaly detection before data reaches production ML workflows.
- Developed layered **dbt** models targeting **BigQuery** to generate observations, evaluate structural rules against historical evidence, and support bounded diagnosis.
- Developing **Python/Airflow** backtesting workflows that tune **adaptive thresholds** from rolling historical baselines across approximately **1 TB of mutation data per 30 days**, producing versioned evidence and alerts.

Twilio San Francisco, CA

Software Engineer Intern *June 2026 – September 2026*

- Built and deployed a **Java/Spring Boot gRPC adapter** that translated Protocol Buffer requests from Conversations Core into Voice-domain HTTP schemas and idempotently created or updated transcription configurations and call filters.
- Validated the service across development and staging with **Mockito unit, Spring Boot integration, and end-to-end tests**, handling 13 requests per minute in staging with response times below **300 ms**.
- Developed a **k6** load-testing framework with runtime Protocol Buffer support to validate **2× baseline traffic**; automated **AWS FIS** latency and traffic experiments with **Terraform**, leading to new dashboards, metrics, logging, and tracing.

University of Chicago Human-Robot Interaction Lab Chicago, IL

Research Assistant *September 2025 – May 2026*

- Spearheaded development of a multimodal **Stretch 3** customization platform for a **69-participant HRI study**, integrating a **Next.js/TypeScript PWA**, Python/Flask services, RealtimeSTT, locally hosted **Qwen2.5-7B**, and **ROS 2**.
- Engineered bidirectional synchronization between voice and tablet inputs, persistent robot configuration, resumable participant sessions, and experiment logging across voice, visual, and multimodal conditions.
- Authored **ROS 2** controllers and motion scripts for navigation, desk wiping, pillow arrangement, and snack retrieval, translating user preferences into room-specific robot behavior.

Microsoft Redmond, WA

Data Analyst Intern *June 2025 – August 2025*

- Optimized **SQL** queries producing **200K+ rows** of global datacenter energy forecasts, reducing execution time by **50%** from 6 to 3 seconds by eliminating redundant joins and duplicate calculations.
- Built a weekly **Python** pipeline that standardized datacenter identifiers and calculated year-over-year and cumulative energy demand, reducing a weeklong Excel workflow to **minutes**; delivered results through a global **Power BI** capacity-planning dashboard.

Gotchu Chicago, IL

Software Engineer Intern *September 2024 – December 2024*

- Built a five-screen onboarding flow for a social food-delivery app using **React Native**, **TypeScript**, and **NativeWind**; integrated Firebase SMS authentication and Firestore-backed user profiles and sessions.
- Developed a reusable **Axios** client with centralized request and error handling for profile submission, contributing to a **TestFlight beta release**.

Projects

Block Blast AI | *GitHub*

- Engineered a **Python** package with a tested 8×8 simulator, linear/MLP/CNN agents, training and evaluation CLIs, model serialization, experiment logs, and an interactive HTML replay renderer.

Conway's Game of Life | *GitHub*

- Built a modular **Haskell** implementation with a purely functional simulation core, interactive Brick TUI, RLE pattern parsing, configurable boundary modes, and CSV analytics.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Haskell, Go, C++, C, Lua
Frameworks: React, Next.js, React Native, Spring Boot, Flask, ROS 2, gRPC, Protocol Buffers, PyTorch
Data/Cloud: BigQuery, dbt, Airflow, Firebase, AWS, Terraform, k6
Developer Tools: Git, Claude Code, Codex, Neovim, tmux